

State-adapted residual elimination for local parameter reliability in inverse physics-informed neural networks

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Abstract

Inverse physics-informed neural networks (PINNs) jointly estimate PDE parameters and a neural representation of the state. This joint training raises a local reliability question: residual loss and raw parameter sensitivity cannot tell whether a parameter perturbation leaves an effect that the network weights cannot reproduce. We derive state-adapted effective parameter signals (SAEPS), a local tangent-space method that removes residual components explainable by neural-state adaptation through a Tikhonov least-squares problem. The method computes state-eliminated curvature, retained sensitivity, and a reduced score from the remaining residual signal. Numerical experiments validate three diagnostic regimes. Manufactured source-amplitude benchmarks are classified as *reliable-local-signal* cases, with retained sensitivities close to one and stable grid, damping, and profile checks. Burgers and coupled reaction–diffusion targets show state absorption: large raw fixed-network sensitivity is mostly removed by local state elimination. Two-temperature targets show score misalignment, where a nonzero retained signal is accompanied by an unfavorable reduced-score direction. Together, these quantities provide a local reduced-residual audit that distinguishes retained, absorbed, and directionally misaligned parameter signals at trained inverse-PINN checkpoints.

Keywords: Inverse problems, Physics-informed neural networks, Parameter reliability, Residual geometry, Tikhonov elimination, Sensitivity analysis

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1. Introduction

Parameter identification for PDE models is a numerical inverse problem in which the state equation, the observation map, and the residual functional jointly determine how a physical parameter can be inferred from data. Classical PDE-constrained formulations make this structure explicit by solving or eliminating the state before analyzing the reduced parameter problem through optimization and sensitivity [1]. General inverse-problem treatments formalize the same reduced-space viewpoint through regularization and parameter-estimation theory [2, 3]. Uncertainty-oriented formulations then add likelihood, posterior, or profile-based summaries for the estimated parameters [4, 5].

Physics-informed neural networks (PINNs) [6] offer an alternative approach. A PINN represents the PDE state by a neural network u_θ with weights θ , and enforces the governing equations, initial and boundary conditions, and data constraints as penalty terms in a loss function. Automatic differentiation computes the PDE residuals directly from the network output, removing the need for a hand-crafted numerical discretization. In a forward setting, only θ is optimized. In an inverse PINN, the physical parameters λ (material coefficients, source amplitudes, etc.) are treated as additional trainable variables and optimized jointly with θ . This allows a PINN to estimate parameters simultaneously with learning the state from sparse and noisy observations; examples include nonlinear diffusion [7], solid mechanics [8], and multiphase poromechanics [9]. Frameworks such as DeepXDE [10] have made these formulations broadly accessible.

Research on PINNs has largely focused on training. Gradient-flow and neural-tangent-kernel analyses explain how loss imbalance and spectral conditioning affect convergence [11, 12]. Empirical studies characterize failure modes caused by stiffness, sampling, or poor initialization [13]. Causal training and adaptive activations modify the optimization dynamics to improve convergence [14, 15]. These contributions address whether a PINN can be trained to a small residual loss; they do not examine whether the resulting parameter estimate is locally supported by residual information after training.

For inverse PINNs specifically, the joint optimization of θ and λ creates a parameter reliability problem that has no direct analogue in classical fixed-solver inverse methods. Because the neural state is an overparameterized, adaptive object, it can locally compensate for changes in the physical

parameters. A parameter perturbation that would produce a clear residual signal under a fixed numerical solver may instead be absorbed by a small adjustment of the network weights. The residual loss can therefore be small while the target parameter remains poorly constrained. We call this mechanism *state absorption*. Parameter-estimation theory with neural-network discretizations acknowledges that the network can act as an implicit regularizer and that the interplay between state approximation and parameter identifiability must be examined at the level of the residual linearization [16, 17].

Several lines of work provide tools for assessing parameter uncertainty, but they do not explicitly address the state-absorption mechanism in inverse PINNs. Bayesian PINNs quantify posterior uncertainty under additional stochastic assumptions [18, 19]. Adversarial and residual-based formulations give complementary uncertainty measures [20]. Profile likelihood distinguishes structural and practical identifiability by reoptimizing nuisance variables over a parameter grid, and it is widely used to test whether a parameter is constrained by data under a given noise model [21]. Sloppiness analysis reveals that nonlinear least-squares models often contain stiff and sloppy parameter combinations, with the local geometry governed by a few dominant directions [22, 23]. Global sensitivity analysis studies variance-based parameter influence [24]. These methods are valuable, but in their standard forms they evaluate identifiability in a space where the state is either fixed or re-solved from scratch. In an inverse PINN, the state is not re-solved; it is a trained network with a high-dimensional tangent space that can absorb parameter-induced residual variation. This motivates a reduced-residual diagnostic that removes the state-explainable component before interpreting parameter sensitivity.

The mathematical operation closest to this removal is reduced-space sensitivity and variable projection. In PDE-constrained optimization, the state is eliminated before the reduced parameter problem is analyzed [25, 26]. In separable nonlinear least squares, variable projection removes a block of variables algebraically and optimizes the remaining parameters [27, 28]. Both approaches form a reduced objective or reduced residual whose curvature reflects parameter information after admissible state adjustments. In an inverse PINN, the neural-state degrees of freedom are not separable in closed form, but they can be eliminated at the level of the trained linearization. SAEPS performs this operation by removing residual components that lie in the span of the network-weight Jacobian through a Tikhonov

least-squares problem. The resulting reduced quadratic model yields state-eliminated curvature, score, retained sensitivity, and a diagnostic step for target-parameter perturbations.

We formulate SAEPS as a residual-space numerical diagnostic for trained inverse-PINN checkpoints. In operational terms, SAEPS takes a trained inverse-PINN solution and a designated target-parameter subset as input. It linearizes the weighted residual with respect to both the network weights θ and the target parameters λ , then removes residual components that lie in the column space of the weight Jacobian J_θ by solving a Tikhonov-regularized least-squares problem. The key object is a residual-space operator $A_\theta^{(\gamma)}$ that maps any residual perturbation to the cost that remains after the best damped neural-state adjustment. Applying this operator to the parameter Jacobian J_λ yields the state-eliminated curvature F_λ^{se} , which measures how much residual signal a parameter perturbation retains after state adaptation. Applying it to the current weighted residual $\bar{r}$ yields the state-eliminated score g_λ^{se} , which indicates the direction of that signal in parameter space. The retained sensitivity η^{se} compares F_λ^{se} with the raw fixed-network curvature, directly quantifying state absorption. None of these quantities requires retraining the network or evaluating the forward model at new parameter values; they are computed entirely from the Jacobians and residuals at the trained solution. The output is a structured local diagnosis: the curvature tells whether any independent signal remains, the score tells whether that signal points toward a useful correction, and the damping-sensitivity scan tells whether the diagnosis is stable under the regularization choice. The formal derivation of $A_\theta^{(\gamma)}$ and the associated diagnostic quantities is given in Section 2.

The contribution of this paper is threefold. First, we formulate local inverse-PINN parameter reliability as a state-eliminated residual-space problem at a trained solution. Second, we derive the SAEPS curvature and score from Tikhonov elimination of neural-state tangent directions. Third, we validate the diagnostics on validity-gated inverse-PINN checkpoints and controlled manufactured benchmarks, showing that SAEPS separates parameter information that is retained, state-absorbed, or score-misaligned. The resulting classification provides information that neither raw fixed-network sensitivity nor final parameter error alone can supply.

2. State-eliminated residual sensitivity for inverse PINNs

2.1. Problem setup and linearized residual geometry

Let u denote the PDE state and $\lambda \in \mathbb{R}^p$ the physical parameters. The inverse problem consists of a governing PDE operator $\mathcal{N}$, initial and boundary constraints $\mathcal{B}$, an observation map $\mathcal{O}$, and data y :

$$\mathcal{N}(u, \lambda) = 0, \quad \mathcal{B}(u, \lambda) = 0, \quad \mathcal{O}(u) = y. \quad (1)$$

An inverse PINN represents the state by a neural network u_θ with weights θ and estimates (θ, λ) jointly by minimizing a weighted residual objective

$$\min_{\theta, \lambda} \frac{1}{2} \|W_d^{1/2} r_d(\theta, \lambda)\|^2 + \frac{1}{2} \|W_p^{1/2} r_p(\theta, \lambda)\|^2 + \frac{1}{2} \|W_b^{1/2} r_b(\theta, \lambda)\|^2, \quad (2)$$

where r_d , r_p , and r_b denote the data, PDE, and boundary/initial-condition residuals with weight matrices W_d , W_p , W_b . Throughout the paper, all Jacobians are taken with respect to the stacked weighted residual

$$\bar{r}(\theta, \lambda) = W^{1/2} r(\theta, \lambda), \quad (3)$$

where r collects the residual blocks and $W^{1/2}$ is the block-diagonal square-root weight matrix.

Linearizing $\bar{r}$ at a trained checkpoint (θ_*, λ_*) yields

$$\delta \bar{r} = J_\theta \delta \theta + J_\lambda \delta \lambda, \quad (4)$$

with Jacobians

$$J_\theta = \frac{\partial \bar{r}}{\partial \theta} \in \mathbb{R}^{m \times n_\theta}, \quad J_\lambda = \frac{\partial \bar{r}}{\partial \lambda} \in \mathbb{R}^{m \times p}, \quad (5)$$

where m is the number of residual evaluations on the diagnostic grid, n_θ the number of network weights, and p the number of target parameters. The raw fixed-network curvature

$$F_\lambda^{\text{raw}} = J_\lambda^\top J_\lambda \quad (6)$$

measures the residual change caused by a parameter perturbation when θ is held fixed.

This fixed-network view is incomplete because the neural state can adapt. If the columns of J_λ lie mainly in the column space of J_θ , then any residual effect of a parameter perturbation $\delta \lambda$ can be approximately reproduced by

a suitable weight perturbation $\delta\theta$. In this case, F_λ^{raw} overstates the amount of independent parameter information retained after state adaptation. We call this phenomenon *state absorption*. To obtain a meaningful local measure of parameter reliability, we must remove residual components that can be explained by the network-weight tangent space and examine what remains.

2.2. Tikhonov state-elimination operator

For a given residual-space vector $y \in \mathbb{R}^m$, we ask how much of y can be explained by the columns of J_θ when large weight perturbations are penalized. This leads to the Tikhonov problem

$$Q_\gamma(y) = \min_{\delta\theta} (\|J_\theta \delta\theta - y\|_2^2 + \gamma \|\delta\theta\|_2^2), \quad \gamma > 0. \quad (7)$$

Equivalently, $Q_\gamma(y)$ asks for the smallest damped residual cost that can be achieved by fitting the perturbation y through an adjustment of the network weights alone. The damping parameter γ stabilizes the elimination when $J_\theta^\top J_\theta$ is ill-conditioned, which is typical for overparameterized neural networks. The minimizer satisfies the normal equations

$$(J_\theta^\top J_\theta + \gamma I) \delta\theta_\gamma = J_\theta^\top y, \quad (8)$$

and substitution back into (7) gives the closed-form quadratic value

$$Q_\gamma(y) = y^\top A_\theta^{(\gamma)} y, \quad (9)$$

where

$$A_\theta^{(\gamma)} = I - J_\theta (J_\theta^\top J_\theta + \gamma I)^{-1} J_\theta^\top. \quad (10)$$

The operator $A_\theta^{(\gamma)}$ is symmetric positive semidefinite for $\gamma > 0$. It maps a residual vector y to the residual cost that remains after the best damped neural-state adjustment has been removed. As $\gamma \rightarrow 0$, $A_\theta^{(\gamma)}$ approaches the orthogonal projector onto the complement of $\text{range}(J_\theta)$ in the Euclidean metric. For ill-conditioned J_θ , the damped version is essential to avoid numerical instability. The entire SAEPS method is built on the action of $A_\theta^{(\gamma)}$ on the parameter Jacobian and the current residual.

2.3. State-eliminated curvature, score, and retained sensitivity

SAEPS derives two complementary quantities from the operator $A_\theta^{(\gamma)}$: one that measures the strength of the residual signal remaining after state elimination, and one that measures its direction.

Curvature. Setting $y = J_\lambda \delta \lambda$ in (9) gives the state-eliminated residual cost for a target-parameter perturbation:

$$Q_\gamma(J_\lambda \delta \lambda) = \delta \lambda^\top F_\lambda^{\text{se}} \delta \lambda, \quad F_\lambda^{\text{se}} = J_\lambda^\top A_\theta^{(\gamma)} J_\lambda. \quad (11)$$

The matrix $F_\lambda^{\text{se}} \in \mathbb{R}^{p \times p}$ is the **state-eliminated curvature**. It measures how much residual signal a parameter perturbation retains after the network weights are allowed to adapt locally.

Score. Applying the same operator to the current weighted residual gives the **state-eliminated score**

$$g_\lambda^{\text{se}} = J_\lambda^\top A_\theta^{(\gamma)} \bar{r}. \quad (12)$$

While F_λ^{se} quantifies the local curvature of the reduced residual model, g_λ^{se} gives the corresponding residual-gradient direction in the same state-eliminated geometry. A target parameter may retain nonzero curvature while the score points in a direction that does not align with the true parameter correction. This separation of strength and direction is central to the diagnostic.

Diagnostic step. From the reduced quadratic model one obtains the local diagnostic step

$$\Delta \lambda^{\text{se}} = -(F_\lambda^{\text{se}} + \rho I)^{-1} g_\lambda^{\text{se}}, \quad \rho \geq 0, \quad (13)$$

where ρ is a parameter-space ridge used when F_λ^{se} is ill-conditioned. When reference parameters are available, the alignment between $\Delta \lambda^{\text{se}}$ and the true correction direction evaluates whether the local reduced score points toward a useful improvement.

Retained sensitivity. To relate the state-eliminated curvature to the raw fixed-network sensitivity, we define the coordinate-wise retained fraction

$$\eta_j^{\text{se}} = \frac{e_j^\top F_\lambda^{\text{se}} e_j}{e_j^\top F_\lambda^{\text{raw}} e_j + \epsilon}, \quad (14)$$

where ϵ guards against division by zero. For multi-parameter targets, direction-wise and trace summaries are used analogously:

$$\eta_v^{\text{se}} = \frac{v^\top F_\lambda^{\text{se}} v}{v^\top F_\lambda^{\text{raw}} v + \epsilon}, \quad \eta_{\text{tr}}^{\text{se}} = \frac{\text{tr}(F_\lambda^{\text{se}})}{\text{tr}(F_\lambda^{\text{raw}}) + \epsilon}. \quad (15)$$

A small retained fraction, $\eta_j^{\text{se}} \ll 1$, indicates state absorption: most of the raw sensitivity is reproducible by neural-state adaptation. A larger η_j^{se} signals that meaningful residual information survives state elimination.

Diagnostic logic. F_λ^{se} and g_λ^{se} are complementary. A parameter may have non-negligible state-eliminated curvature but a score that points away from the reference correction—a situation we call *score misalignment*—or it may have both weak curvature and an unhelpful score direction. Retained sensitivity η^{se} normalizes the curvature and facilitates comparison across parameters and benchmarks. A retained fraction close to one is therefore a necessary but not sufficient indicator of local parameter reliability: the reduced score g_λ^{se} may still point in an unhelpful direction, producing a score-misaligned case. Together, these quantities provide a structured local picture: the curvature tells whether any signal remains after state adaptation, and the score tells whether that signal supports a useful correction.

2.4. Reduced-space interpretation

The preceding quantities are the exact gradient and curvature of a locally reduced least-squares model, rather than independent diagnostic constructions. At a fixed checkpoint, consider the regularized quadratic model

$$\Phi_\gamma(\delta\theta, \delta\lambda) = \frac{1}{2} \|\bar{r} + J_\theta\delta\theta + J_\lambda\delta\lambda\|_2^2 + \frac{\gamma}{2} \|\delta\theta\|_2^2, \quad (16)$$

and eliminate the network-weight increment by defining

$$\hat{\Phi}_\gamma(\delta\lambda) = \min_{\delta\theta} \Phi_\gamma(\delta\theta, \delta\lambda). \quad (17)$$

Proposition 1 (Local reduced gradient and curvature). *For every $\gamma > 0$, the reduced objective in (17) is*

$$\hat{\Phi}_\gamma(\delta\lambda) = \frac{1}{2} (\bar{r} + J_\lambda\delta\lambda)^\top A_\theta^{(\gamma)} (\bar{r} + J_\lambda\delta\lambda). \quad (18)$$

Consequently,

$$\nabla \hat{\Phi}_\gamma(0) = g_\lambda^{\text{se}}, \quad \nabla^2 \hat{\Phi}_\gamma(\delta\lambda) = F_\lambda^{\text{se}}. \quad (19)$$

Proof. Set $x = \bar{r} + J_\lambda\delta\lambda$. The first-order condition for minimizing (16) with respect to $\delta\theta$ gives

$$\delta\theta_\gamma(\delta\lambda) = -(J_\theta^\top J_\theta + \gamma I)^{-1} J_\theta^\top x.$$

Substitution into (16) yields (18). Differentiating the resulting quadratic with respect to $\delta\lambda$ gives (19) together with the definitions in (12) and (11). $\square$

Equivalently, F_λ^{se} is the Schur complement of the regularized network-weight block in the joint Gauss–Newton matrix

$$\begin{bmatrix} J_\theta^\top J_\theta + \gamma I & J_\theta^\top J_\lambda \\ J_\lambda^\top J_\theta & J_\lambda^\top J_\lambda \end{bmatrix}. \quad (20)$$

Indeed,

$$F_\lambda^{\text{se}} = J_\lambda^\top J_\lambda - J_\lambda^\top J_\theta (J_\theta^\top J_\theta + \gamma I)^{-1} J_\theta^\top J_\lambda. \quad (21)$$

As $\gamma \downarrow 0$, this expression approaches

$$J_\lambda^\top [I - J_\theta (J_\theta^\top J_\theta)^\dagger J_\theta^\top] J_\lambda, \quad (22)$$

which is the Gauss–Newton curvature after orthogonal elimination of the network-state tangent directions. SAEPS can therefore be interpreted as a Tikhonov-regularized local profiling of the linearized least-squares residual. Under an independent Gaussian residual model, the same reduced quadratic is the local approximation obtained by profiling the nuisance state coordinates.

2.5. Matrix-free implementation and practical notes

The matrix $A_\theta^{(\gamma)}$ is never formed explicitly. Its action on a vector y is computed by solving the regularized normal system

$$(J_\theta^\top J_\theta + \gamma I)z = J_\theta^\top y \quad (23)$$

and returning $A_\theta^{(\gamma)}y = y - J_\theta z$. The required operations are Jacobian-vector products $v \mapsto J_\theta v$, vector-Jacobian products $w \mapsto J_\theta^\top w$, and iterative solves with the regularized normal matrix. Conjugate gradients are used with a predefined tolerance and maximum iteration count; the convergence flag and iteration count are recorded for every solve.

When the target dimension is small, the columns of J_λ can be stored explicitly; for larger targets, J_λ is applied matrix-free. The damping is scaled relative to the state-Jacobian spectrum:

$$\gamma = \gamma_\alpha \lambda_{\max}(J_\theta^\top J_\theta), \quad (24)$$

where $\lambda_{\max}$ is estimated by a few power iterations and γ_α is a dimensionless grid declared in the experimental configuration. Because the norm $\|\delta\theta\|$ depends on the network parameterization, each SAEPS result must be reported together with the network architecture and the γ grid. The effect of the damping

Algorithm 1 SAEPS at a validity-passing checkpoint

Require: checkpoint (θ_*, λ_*) , weighted residual $\bar{r}$, target parameter subset, damping grid $\{\gamma_\alpha\}$, parameter ridge ρ , diagnostic grid, residual weights, parameter coordinates.

- 1: Verify the checkpoint-validity gate and record PDE, data, boundary, initial-condition, and state residuals.
- 2: Record network architecture, activation, parameter scaling, and checkpoint metadata.
- 3: Compute Jacobian actions J_θ , $J_\theta^\top$, and J_λ by automatic differentiation on the diagnostic grid.
- 4: **for** each γ_α in the declared grid **do**
- 5: Set $\gamma = \gamma_\alpha \lambda_{\max}(J_\theta^\top J_\theta)$.
- 6: For any required y , solve $(J_\theta^\top J_\theta + \gamma I)z = J_\theta^\top y$ via conjugate gradients; return $A_\theta^{(\gamma)}y = y - J_\theta z$.
- 7: Record Krylov iterations, residual norm, condition estimate, and convergence flag.
- 8: Assemble $F_\lambda^{\text{se}} = J_\lambda^\top A_\theta^{(\gamma)} J_\lambda$ and $g_\lambda^{\text{se}} = J_\lambda^\top A_\theta^{(\gamma)} \bar{r}$.
- 9: Compute η^{se} and $\Delta\lambda^{\text{se}} = -(F_\lambda^{\text{se}} + \rho I)^{-1} g_\lambda^{\text{se}}$.
- 10: If reference parameters are available, record score alignment and harmful-step indicators.
- 11: **end for**
- 12: Return $\{F_\lambda^{\text{se}}, g_\lambda^{\text{se}}, \eta^{\text{se}}, \Delta\lambda^{\text{se}}\}$, damping metadata, validity metadata, and provenance.

choice on the qualitative diagnosis is assessed through a damping-sensitivity scan over the declared γ_α values.

SAEPS is applied only to checkpoints that pass a predefined validity gate on the PDE, data, boundary/initial-condition, and state residuals. This gate screens out checkpoints for which state, residual-block, or linear-solve errors would make the diagnostic uninterpretable. Checkpoints that fail the gate are reported separately and excluded from the main diagnostic aggregation.

3. Numerical experiments

The numerical study tests whether a target-parameter residual signature is retained after local neural-state adaptation, absorbed by the network-weight tangent space, or accompanied by a reduced-score direction that is inconsistent

with the reference correction. The experiments are organized into retained-signal positive controls, state-absorption tests, and score-misalignment tests. All trainable physical parameters are represented in natural-log coordinates. Complete manufactured formulas, source constructions, sampling rules, and row-level validity values are reported in the Supplementary Material.

3.1. Benchmark definitions

Benchmark	State equation and domain	Reference / observations	Target coordinates
Scalar source amplitude	$u_t - Du_{xx} + cu = \lambda q + s$ on $[0, 1] \times [0, 1]$, with $D = 0.05$ and $c = 1$	Closed-form manufactured state; observations, initial data, and boundary data sampled from u^*	$\log \lambda, \lambda^* = 1.0, \lambda_0 = 0.98$
Coupled source amplitude	$u_t = D_u u_{xx} - au + bv + \lambda_u q_u + s_u$, $v_t = D_v v_{xx} + au - bv + \lambda_v q_v + s_v$ on $[0, 1] \times [0, 1]$	Closed-form manufactured two-channel state; both channels observed	$\log \lambda_u, \log \lambda_v$, with $(\lambda_u^*, \lambda_v^*) = (1.0, 0.8)$
Burgers	$u_t + uu_x = \mu u_{xx}$ on $[0, 1] \times [0, 0.4]$ with periodic boundary condition and $u(x, 0) = \sin(2\pi x)$	Periodic finite-difference reference on a 128×2001 grid; data obtained by periodic spatial indexing and linear interpolation in time	$\log \mu, \mu^* = 0.05, \mu_0 = 0.08$
Coupled reaction-diffusion	$u_t = D_u u_{xx} - au + bv + s_u$, $v_t = D_v v_{xx} + au - bv + s_v$ on $[0, 1] \times [0, 0.4]$, with $D_u = 0.03$, $D_v = 0.02$	Manufactured states $u^* = 0.8 + 0.2 \sin(\pi x)e^{-t}$ and $v^* = 0.6 + 0.15 \cos(\pi x)e^{-0.7t}$; both channels observed	$\log a, \log b$, with $(a^*, b^*) = (0.8, 0.75)$ and initial values $(0.6, 0.95)$
Two-temperature	$u_{e,t} = \kappa_e u_{e,xx} - G(u_e - u_l) + I_0 S$, $u_{l,t} = G(u_e - u_l)$ on $[0, 1] \times [0, 0.4]$, with $c_e = c_l = 1$	Manufactured states $u_l^* = 0.4 + 0.15 \sin(\pi x)e^{-0.5t}$ and $u_e^* = u_l^* + 0.5\partial_t u_l^*$; both channels observed	$\log I_0, \log \kappa_e, \log G$, with $(I_0^*, \kappa_e^*, G^*) = (1.0, 0.4, 2.0)$

Table 1: Benchmark definitions used for the main SAEPS validation. The complete forcing terms s , s_u , s_v , and $S(x, t)$ are obtained by substituting the stated reference states into the corresponding governing equations and are listed in the Supplementary Material.

Table 1 summarizes the benchmark matrix used in the main text. The source-amplitude benchmarks are controlled positive cases in which

a target parameter multiplies a prescribed source signature. Burgers and coupled reaction–diffusion test whether large fixed-network sensitivity survives state adaptation. The two-temperature model tests whether nonzero state-eliminated curvature is accompanied by a useful reduced-score direction.

3.2. Training and diagnostic protocol

For the Burgers, coupled reaction–diffusion, and two-temperature chain, the inverse PINN is a tanh multilayer perceptron with width 24 and depth 3. Adam is run for 600 epochs with learning rate 2×10^{-3} . Each epoch uses $n_{\text{PDE}} = 64$, $n_{\text{data}} = 96$, $n_{\text{IC}} = 48$, and $n_{\text{BC}} = 48$ training samples. Interior PDE and data points are sampled uniformly in the corresponding space–time domain; initial-condition points are sampled uniformly in x at $t = 0$; boundary points are sampled uniformly in t at $x = 0$ and $x = 1$. The sampler is seed controlled and resamples every epoch. The main validation uses noise level $\sigma = 0$ and observation fraction 1.0. The training loss uses block weights $w_{\text{PDE}} = 5$, $w_{\text{data}} = 10$, $w_{\text{IC}} = 10$, and $w_{\text{BC}} = 2$.

The SAEPS diagnostic grid for the main chain uses $n_{\text{PDE}} = 16$, $n_{\text{data}} = 24$, $n_{\text{IC}} = 16$, and $n_{\text{BC}} = 16$ samples drawn once per checkpoint from the same distributions. The weighted residual used by SAEPS applies the same block weights as the training loss. The core diagnostics use

$$\gamma = \gamma_{\alpha} \lambda_{\max}(J_{\theta}^{\top} J_{\theta}), \quad \gamma_{\alpha} = 10^{-7}, \quad (25)$$

and the damping audit uses the extended grid

$$\gamma_{\alpha} \in \{10^{-12}, 10^{-10}, 10^{-8}, 10^{-6}, 10^{-4}, 10^{-2}, 1, 10^2\}$$

for consistency checks. The target-level values in Table 2 are evaluated at the core scale $\gamma_{\alpha} = 10^{-7}$. The conjugate-gradient tolerance is 10^{-8} with a maximum of 300 iterations for the main chain. Source-amplitude positive controls use the same residual-block structure with larger training samples and a maximum of 500 conjugate-gradient iterations; details are provided in the Supplementary Material.

3.3. Validity gate and target-label rules

A checkpoint from the main three-benchmark chain enters target-level aggregation only if it passes the checkpoint-quality gate

$$\begin{aligned} \text{PASS} = \mathbf{1}\{ & E_{\text{PDE}} \leq 5 \times 10^{-2} \wedge E_{\text{data}} \leq 5 \times 10^{-2} \wedge E_{\text{IC}} \leq 5 \times 10^{-2} \\ & \wedge E_{\text{BC}} \leq 5 \times 10^{-2} \wedge \text{nRMSE}_{\text{state}} \leq 5 \times 10^{-1}\}. \end{aligned} \quad (26)$$

The gate is applied to unweighted residual-block MSEs. State nRMSE is computed over all state channels as

$$\text{nRMSE}_{\text{state}} = \frac{\|u_\theta - u_{\text{ref}}\|_2}{\|u_{\text{ref}}\|_2 + 10^{-12}}, \quad (27)$$

where the tensor norm concatenates all channels. The SAEPS diagnostic solve gate is separate: every conjugate-gradient solve required to apply $A_\theta^{(\gamma)}$ to the columns of J_λ and to the residual vector must satisfy

$$\frac{\|(J_\theta^\top J_\theta + \gamma I)z - J_\theta^\top y\|_2}{\|J_\theta^\top y\|_2 + 10^{-30}} \leq 10^{-8}. \quad (28)$$

Rows failing either gate are excluded from the main target-level medians and are reported separately. In the main three-benchmark chain, Burgers contributes four validity-passing rows; coupled reaction–diffusion and two-temperature contribute five validity-passing rows per target.

When reference parameters are available, the validation-only score alignment is

$$\text{align} = \frac{\langle \Delta\lambda^{\text{se}}, \lambda_{\text{ref}} - \lambda_{\text{ckpt}} \rangle}{\|\Delta\lambda^{\text{se}}\|_2 \|\lambda_{\text{ref}} - \lambda_{\text{ckpt}}\|_2 + 10^{-30}}. \quad (29)$$

This quantity is used only for manufactured or synthetic validation because it requires a reference parameter. The weak-eigendirection diagnostic is the smallest eigenvalue of the symmetrized state-eliminated curvature,

$$\lambda_{\min}^{\text{weak}} = \lambda_{\min} \left(\frac{F_\lambda^{\text{se}} + (F_\lambda^{\text{se}})^\top}{2} \right). \quad (30)$$

The internal target decision rule uses the priority order

$$\begin{cases} \textit{score-misaligned}, & \text{align} < 0, \\ \textit{state-absorbed}, & \text{align} \geq 0 \text{ and } \eta^{\text{se}} < 0.15, \\ \textit{weak-eigendirection}, & \text{align} \geq 0, \eta^{\text{se}} \geq 0.15, \lambda_{\min}^{\text{weak}} < 10^{-4}, \\ \textit{locally reliable}, & \text{otherwise.} \end{cases} \quad (31)$$

The paper-facing summary reports the three observed regimes *reliable-local-signal*, *state-absorbed*, and *score-misaligned*. Grid, damping, and profile audits are reported as consistency checks rather than as replacement labels.

4. Results and discussion

4.1. Benchmark matrix and diagnostic classification

The benchmark matrix in Section 3 is designed to test three SAEPS regimes: retained local signal, state absorption, and score misalignment. Table 2 summarizes the target-level classification obtained after applying SAEPS to validity-passing checkpoints. All scalar values in the table are medians over validity-passing checkpoints unless otherwise stated.

Benchmark / target	SAEPS evidence and consistency checks	Classification
<i>Retained local signal</i>		
scalar source amplitude / $\log \lambda$	$\eta^{\text{se}} = 0.995162$; score alignment = 1. Checks: grid-stable; plateau-stable; checkpoint-centered.	<i>reliable-local-signal</i>
coupled source amplitude / $\log \lambda_u$	$\eta^{\text{se}} = 0.995788$; score alignment = 1. Checks: grid-stable; plateau-stable; checkpoint-centered.	<i>reliable-local-signal</i>
coupled source amplitude / $\log \lambda_v$	$\eta^{\text{se}} = 0.999656$; score alignment = 1. Checks: grid-stable; plateau-stable; checkpoint-centered.	<i>reliable-local-signal</i>
<i>State absorption</i>		
Burgers / $\log \mu$	$F_{\lambda}^{\text{raw}} = 42770.6$; $\eta^{\text{se}} = 0.032039$. Checks: grid-stable; damping-sensitive.	<i>state-absorbed</i>
coupled reaction–diffusion / $\log a$	$F_{\lambda}^{\text{raw}} = 19767.7$; $\eta^{\text{se}} = 0.068481$. Checks: grid-stable; damping-sensitive.	<i>state-absorbed</i>
coupled reaction–diffusion / $\log b$	$F_{\lambda}^{\text{raw}} = 7740.64$; $\eta^{\text{se}} = 0.068123$. Checks: grid-stable; damping-sensitive.	<i>state-absorbed</i>
<i>Score misalignment</i>		
two-temperature / $\log G$	$\eta^{\text{se}} = 0.046764$; score alignment = -0.167709 . Checks: grid-sensitive; damping-sensitive.	<i>score-misaligned</i>
two-temperature / $\log I_0$	$\eta^{\text{se}} = 0.044639$; score alignment = -0.167709 . Checks: grid-sensitive; damping-sensitive.	<i>score-misaligned</i>
two-temperature / $\log \kappa_e$	$\eta^{\text{se}} = 0.041688$; score alignment = -0.167709 . Checks: grid-sensitive; damping-sensitive.	<i>score-misaligned</i>

Table 2: Target-level SAEPS diagnostic classification across retained, state-absorbed, and score-misaligned regimes. Reported scalar values are medians over validity-passing checkpoints unless otherwise stated.

Table 2 and Fig. 1 show that SAEPS separates the tested targets into three interpretable regimes. The manufactured source-amplitude targets retain nearly all of their parameter residual signal after state elimination and satisfy all consistency checks. Burgers and coupled reaction–diffusion have large raw sensitivity but small retained fractions, indicating state absorption. The two-

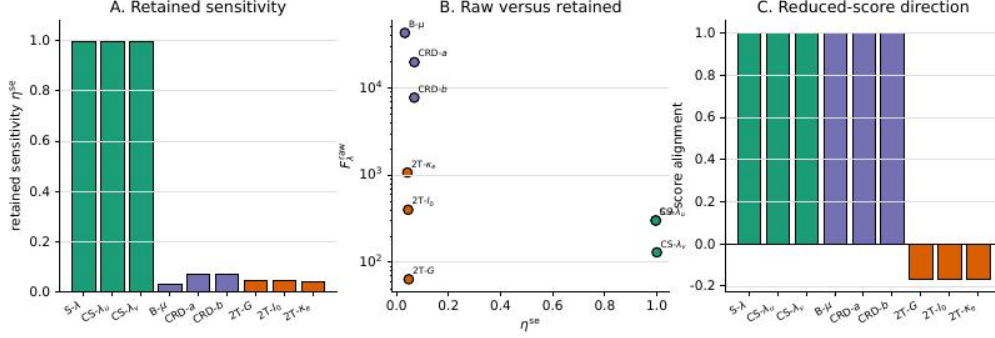


Figure 1: Three SAEPS diagnostic regimes across target rows. Manufactured source-amplitude targets retain nearly all parameter residual signal after state elimination. Burgers and coupled reaction–diffusion have large raw fixed-network sensitivity but low retained sensitivity, indicating state absorption. Two-temperature targets show unfavorable reduced-score alignment, indicating score misalignment. Reported values are medians over validity-passing checkpoints. Axis abbreviations denote scalar source amplitude (S), coupled source amplitude (CS), Burgers (B), coupled reaction–diffusion (CRD), and two-temperature (2T).

temperature targets have a nonzero retained signal but an unfavorable reduced-score direction, indicating score misalignment. The following subsections examine the regimes and the consistency checks supporting the classifications. The complete target-level numerical summary is reported in Supplementary Table S4.

4.2. Retained local signals and reduced-residual interpretation

The manufactured source-amplitude benchmarks serve as the positive-control group. In these problems the target parameter multiplies a prescribed source term whose empirical residual signature is designed to be transverse to the neural-state tangent response at the trained solution. Consequently, the columns of J_λ retain a large component after the Tikhonov elimination of $\text{range}(J_\theta)$, and the state-eliminated curvature remains close to the raw fixed-network sensitivity.

All three source-amplitude targets are classified as *reliable-local-signal*. The scalar source-amplitude target $\log \lambda$ reaches a median state nRMSE of 2.54×10^{-5} and a median parameter log-error of 6.76×10^{-6} , with a retained sensitivity $\eta^{se} = 0.995162$ and a score alignment of 1. The coupled source-amplitude targets behave similarly: for $\log \lambda_u$ the median state nRMSE is 2.72×10^{-5} , the median parameter log-error is 5.95×10^{-6} , and $\eta^{se} =$

0.995788; for $\log \lambda_v$ the median parameter log-error is 4.21×10^{-6} and $\eta^{\text{se}} = 0.999656$. All three targets are grid-stable, plateau-stable under the damping scan, and checkpoint-centered in the profile audit.

These results directly reflect the reduced-residual logic of SAEPS. When the parameter residual direction lies almost entirely outside the neural-state tangent response, the operator $A_\theta^{(\gamma)}$ leaves it essentially unchanged, so $F_\lambda^{\text{se}} \approx F_\lambda^{\text{raw}}$. The near-unit retained sensitivity and positive score alignment are consistent with a strong and correctly oriented reduced residual quadratic model. In such cases, the SAEPS diagnostic coincides with what a classical reduced-parameter sensitivity analysis would report if the state were eliminated exactly.

4.3. State absorption and score misalignment: mechanisms and diagnostic value

The two cautionary regimes demonstrate why raw fixed-network sensitivity alone is insufficient for inverse-PINN reliability and how SAEPS separates distinct failure modes.

State absorption despite large raw sensitivity. The Burgers and coupled reaction–diffusion targets exhibit large raw curvature but small retained fractions. For Burgers/ $\log \mu$, $F_\lambda^{\text{raw}} = 42770.6$ and $\eta^{\text{se}} = 0.032039$; for coupled reaction–diffusion/ $\log a$, $F_\lambda^{\text{raw}} = 19767.7$ and $\eta^{\text{se}} = 0.068481$; for $\log b$, $F_\lambda^{\text{raw}} = 7740.64$ and $\eta^{\text{se}} = 0.068123$. The columns of J_λ lie predominantly in the column space of J_θ , so the damped projection $A_\theta^{(\gamma)} J_\lambda$ yields only a weak residual signal. The targets are therefore classified as *state-absorbed*.

This regime reveals a mechanism that is not visible in fixed-network Fisher information. A large F_λ^{raw} measures the effect of a parameter perturbation under frozen network weights. SAEPS shows that, once the weights are allowed to adapt locally, the same residual perturbation can be reproduced almost entirely by a small $\delta\theta$, leaving little independent parameter information. The *state-absorbed* label thus prevents overinterpretation of a numerically large but geometrically redundant sensitivity.

Score misalignment with non-negligible reduced curvature. The two-temperature targets $\log G$, $\log I_0$, and $\log \kappa_e$ show a different cautionary pattern. The retained sensitivity is nonzero but small ($\eta^{\text{se}} = 0.046764$, 0.044639 , 0.041688), and the reduced-score direction is unfavorable, with a median score alignment of -0.167709 for all three targets. The diagnosis is *score-misaligned*.

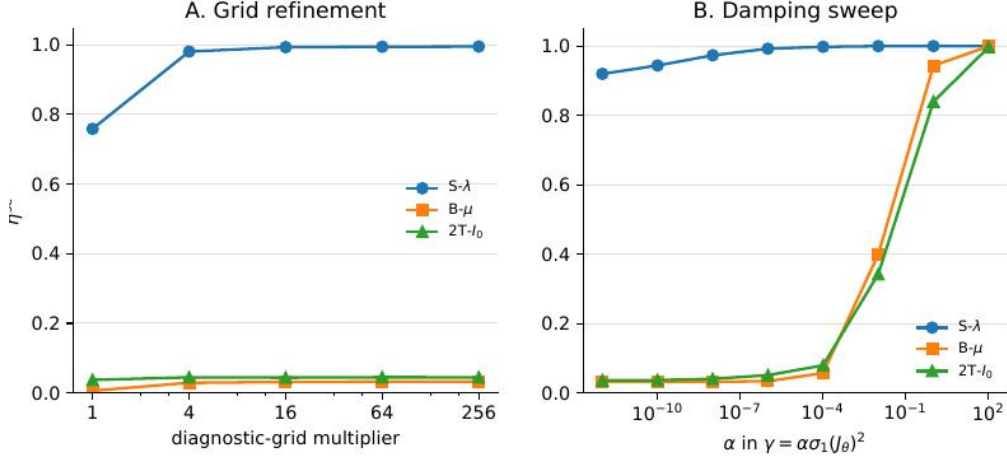


Figure 2: Grid and damping consistency checks for representative target rows. The source-amplitude target remains stable under diagnostic-grid refinement and damping variation, whereas cautionary targets expose either weak retained signal or damping-sensitive behavior. Full grid/rank and damping traces are reported in the Supplementary Material.

This separation between curvature strength and score direction is central to the SAEPS logic. A non-negligible F_λ^{se} indicates that some residual signal survives state elimination, but the signal may point in a direction that does not correspond to a useful parameter correction. The two-temperature results show that even when curvature-based evidence is present, the local residual gradient can still be inconsistent with the true parameter displacement. SAEPS captures this misalignment through the reduced score, providing a diagnostic that curvature alone cannot offer.

4.4. Consistency checks and stress tests

Every SAEPS classification is accompanied by numerical consistency checks that make the label depend on grid, damping, and profile behavior rather than on a single favorable scalar.

Grid and rank audit. The diagnostic residual grid is refined through the multipliers $\{1, 4, 16, 64, 256\}$ relative to the base diagnostic grid. A target is treated as grid-stable when its classification and retained-sensitivity ordering are preserved at the largest grid levels. All source-amplitude and state-absorbed targets are grid-stable; the two-temperature targets are grid-sensitive, indicating that their residual geometry has not yet converged at the base resolution.

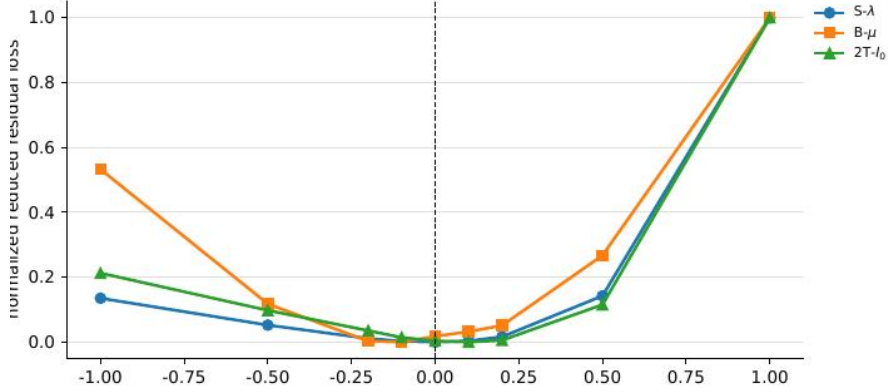


Figure 3: Reduced profile audit for representative target rows. The source-amplitude target is checkpoint-centered and passes the other consistency checks. For state-absorbed and score-misaligned targets, profile behavior alone does not override the cautionary classification because retained sensitivity, damping stability, or reduced-score direction fails the declared reliability criteria.

Damping audit. The Tikhonov parameter is written as $\gamma = \gamma_\alpha \sigma_1(J_\theta)^2$, where $\sigma_1(J_\theta)$ is the largest singular value of the state Jacobian. The damping audit uses the extended grid $\gamma_\alpha \in \{10^{-12}, 10^{-10}, 10^{-8}, 10^{-6}, 10^{-4}, 10^{-2}, 1, 10^2\}$, while the target-level medians in Table 2 are evaluated at the core scale $\gamma_\alpha = 10^{-7}$. A target is considered plateau-stable when the retained sensitivity remains high and approximately constant across the available sweep, without a qualitative change in the diagnostic label under the median-based criterion. Source-amplitude targets are plateau-stable; state-absorbed and score-misaligned targets show damping sensitivity. This sensitivity is itself a diagnostic: a label that depends strongly on γ signals a target whose residual signal is fragile at the chosen linearization scale.

Profile audit. A quadratic reduced-profile scan along the diagnostic step $\Delta\lambda^{\text{se}}$ checks whether the trained parameter lies near the minimum of the reduced residual model. Source-amplitude targets are checkpoint-centered and pass the other consistency checks. For the state-absorbed and score-misaligned targets, the profile audit does not override the cautionary classification, because the retained fraction, damping stability, or reduced-score direction fails the declared reliability criteria.

Stress tests. Selected checks on observation noise, data sparsity, and

network architecture for Burgers/ $\log \mu$ and coupled reaction–diffusion/ $\log a$ support the *state-absorbed* interpretation for these selected targets under observation-noise, sparsity, and architecture perturbations. Full target-level stress-test details are reported in the Supplementary Material.

Collectively, Fig. 2 and Fig. 3 show how the consistency checks prevent a *reliable-local-signal* label unless retained sensitivity is high, the damping plateau is flat, the grid convergence is clean, and the profile is centered. The checks thus operationalize the SAEPS diagnostic logic as a structured audit rather than a single-number threshold. Full grid/rank, damping, and profile audit traces are provided in Supplementary Figs. S2–S4, and the selected noise/sparsity and architecture stress tests are reported in Supplementary Figs. S5–S6.

4.5. Relation to classical sensitivity concepts, scope, and limitations

The SAEPS formulation places it within a family of reduced-space sensitivity methods, but its scope is deliberately local and tied to a fixed trained solution.

Relation to profile likelihood and variable projection. Profile likelihood evaluates parameter identifiability by fixing a parameter and reoptimizing all nuisance variables; the resulting profile reveals whether the data constrain that parameter. SAEPS performs a quadratic local analog of this reoptimization: it removes residual directions that can be absorbed by the network weights through a single Tikhonov solve, without iterative retraining. In the limiting case in which the state is eliminated exactly and the objective is linear least squares, F_λ^{se} reduces to the Schur-complement curvature for the target parameter. SAEPS extends this reduced-space logic to overparameterized neural-state models by introducing the regularized tangent-space projector $A_\theta^{(\gamma)}$.

Raw Fisher information versus state-eliminated curvature. Raw fixed-network sensitivity $F_\lambda^{\text{raw}} = J_\lambda^\top J_\lambda$ measures the residual change when the network is frozen. SAEPS replaces it with $F_\lambda^{\text{se}} = J_\lambda^\top A_\theta^{(\gamma)} J_\lambda$, which measures the residual change that remains after admissible neural-state adaptation. The difference between the two is the state-absorption effect, quantified by the retained sensitivity η^{se} . In the benchmarks, this difference is negligible for the source-amplitude targets ($\eta^{\text{se}} \approx 1$) but dramatic for Burgers ($\eta^{\text{se}} \approx 0.03$), illustrating that raw sensitivity can overstate local parameter information by more than an order of magnitude when the neural trial space is flexible.

Scope and limitations. SAEPS audits the trained residual linearization rather than the global inverse problem. Its diagnostic statements are tied to the trained solution, the residual weighting, the neural architecture, and the declared Tikhonov scale γ . Score alignment is used only for manufactured-benchmark validation because it requires reference parameters; in real inverse problems the deployable evidence is the combination of state-eliminated curvature, retained sensitivity, and the grid, damping, and profile consistency checks. This role is complementary to global identifiability analysis, posterior uncertainty quantification, and cross-validation with classical solvers.

5. Conclusion

This paper has introduced SAEPS, a local reduced-residual diagnostic for inverse physics-informed neural networks. The method eliminates residual components that can be explained by the network-weight tangent response and reports state-eliminated curvature, retained sensitivity, and reduced-score quantities for the target parameters.

The numerical validation confirms three diagnostic regimes predicted by the construction. Manufactured source-amplitude problems retain most of their parameter signal and show stable grid, damping, and profile checks. Burgers and coupled reaction–diffusion exhibit state absorption, where large raw fixed-network sensitivity is mostly removed by local state elimination. The two-temperature benchmark exhibits score misalignment, where nonzero retained residual signal is accompanied by an unfavorable reduced-score direction.

These results support SAEPS as a local reliability audit for trained inverse-PINN solutions. Its role is complementary: SAEPS tests whether the trained residual linearization supports, absorbs, or misaligns a target-parameter signal, whereas global identifiability analysis and posterior uncertainty quantification address different levels of inverse-problem uncertainty.

Supplementary material

Reproducibility trace

The source-amplitude positive controls, the Burgers, coupled reaction–diffusion, and two-temperature checkpoints, and all paper-facing figures and table mirrors are generated from the current paper package. The corresponding provenance scripts are listed by role: source-amplitude positive controls in `outputs/saeps_positive_controls_v1/`; main-chain inverse-PINN checkpoints in `experiments/`; paper figures and table mirrors in `scripts/` and `outputs/paper_figures_v1/plot_data/`. The copied submission figures are stored in `figures/`.

Appendix A. Complete Benchmark Definitions

A.1 Scalar Source-Amplitude Positive Control

The scalar source-amplitude benchmark is posed on $(x, t) \in [0, 1] \times [0, 1]$:

$$u_t - Du_{xx} + cu = \lambda q(x, t) + s(x, t), \quad D = 0.05, \quad c = 1. \quad (32)$$

The trainable target coordinate is $\log \lambda$, with $\lambda^* = 1.0$ and initial value $\lambda_0 = 0.98$. The manufactured state and source signature are

$$u^*(x, t) = \sin(\pi x)e^{-t} + 0.2 \sin(3\pi x) \sin(2\pi t), \quad (33)$$

$$q(x, t) = \sin(2\pi x)(1 + t). \quad (34)$$

The forcing is

$$s(x, t) = u_t^* - Du_{xx}^* + cu^* - \lambda^* q(x, t). \quad (35)$$

Initial data, boundary data, and observations are sampled from u^* .

A.2 Coupled Source-Amplitude Positive Control

The coupled source-amplitude benchmark is posed on $(x, t) \in [0, 1] \times [0, 1]$:

$$u_t = D_u u_{xx} - au + bv + \lambda_u q_u(x, t) + s_u(x, t), \quad (36)$$

$$v_t = D_v v_{xx} + au - bv + \lambda_v q_v(x, t) + s_v(x, t), \quad (37)$$

with fixed coefficients

$$D_u = 0.04, \quad D_v = 0.02, \quad a = 0.8, \quad b = 0.75. \quad (38)$$

The trainable coordinates are $\log \lambda_u$ and $\log \lambda_v$, with $(\lambda_u^*, \lambda_v^*) = (1.0, 0.8)$ and initial values $(0.98, 0.82)$. The manufactured states and source signatures are

$$u^*(x, t) = \sin(\pi x)e^{-t} + 0.1 \sin(2\pi x) \sin t, \quad (39)$$

$$v^*(x, t) = \cos(0.5\pi x)e^{-0.5t} + 0.1 \sin(3\pi x) \cos t, \quad (40)$$

$$q_u(x, t) = \sin(2\pi x)(1 + t), \quad (41)$$

$$q_v(x, t) = \cos(3\pi x)(1 + 0.5t). \quad (42)$$

The forcing terms are

$$s_u(x, t) = u_t^* - D_u u_{xx}^* + au^* - bv^* - \lambda_u^* q_u(x, t), \quad (43)$$

$$s_v(x, t) = v_t^* - D_v v_{xx}^* - au^* + bv^* - \lambda_v^* q_v(x, t). \quad (44)$$

Both state channels are observed, and initial and boundary data are sampled from (u^*, v^*) .

A.3 Burgers Benchmark

The Burgers benchmark uses

$$u_t + uu_x = \mu u_{xx}, \quad x \in [0, 1], \quad t \in [0, 0.4], \quad (45)$$

with periodic boundary condition $u(0, t) = u(1, t)$. The trainable coordinate is $\log \mu$, with $\mu^* = 0.05$ and initial value $\mu_0 = 0.08$. The finite-difference reference uses

$$\begin{aligned} u(x, 0) &= \sin(2\pi x), & N_x &= 128, & N_t &= 2001, \\ \Delta x &= 1/128, & \Delta t &= 0.4/2000. \end{aligned} \quad (46)$$

The time update is explicit Euler with centered periodic finite differences,

$$u^{k+1} = u^k + \Delta t [-u^k D_x u^k + \mu D_{xx} u^k], \quad (47)$$

where D_x and D_{xx} are implemented with periodic array shifts. PINN data values are obtained by periodic nearest-neighbor indexing in space and linear interpolation in time. No source term is used in this benchmark.

A.4 Coupled Reaction–Diffusion Benchmark

The coupled reaction–diffusion benchmark uses

$$u_t = D_u u_{xx} - au + bv + s_u(x, t), \quad (48)$$

$$v_t = D_v v_{xx} + au - bv + s_v(x, t), \quad (49)$$

on $[0, 1] \times [0, 0.4]$, with $D_u = 0.03$ and $D_v = 0.02$. The trainable coordinates are $\log a$ and $\log b$, with $(a^*, b^*) = (0.8, 0.75)$ and initial values $(0.6, 0.95)$. The manufactured states are

$$u^*(x, t) = 0.8 + 0.2 \sin(\pi x) e^{-t}, \quad (50)$$

$$v^*(x, t) = 0.6 + 0.15 \cos(\pi x) e^{-0.7t}. \quad (51)$$

The source functions are

$$s_u(x, t) = u_t^* - D_u u_{xx}^* + a^* u^* - b^* v^*, \quad (52)$$

$$s_v(x, t) = v_t^* - D_v v_{xx}^* - a^* u^* + b^* v^*. \quad (53)$$

Initial conditions, Dirichlet boundary conditions at $x = 0$ and $x = 1$, and observations are sampled from (u^*, v^*) .

A.5 Two-Temperature Benchmark

The two-temperature benchmark is implemented in normalized heat-capacity units:

$$u_{e,t} = \kappa_e u_{e,xx} - G(u_e - u_l) + I_0 S(x, t), \quad (54)$$

$$u_{l,t} = G(u_e - u_l), \quad (55)$$

on $[0, 1] \times [0, 0.4]$. Thus the residual code corresponds to $c_e = c_l = 1$. The trainable coordinates are $\log I_0$, $\log \kappa_e$, and $\log G$, with $(I_0^*, \kappa_e^*, G^*) = (1.0, 0.4, 2.0)$ and initial values $(0.8, 0.55, 1.6)$. The manufactured states are

$$u_l^*(x, t) = 0.4 + 0.15 \sin(\pi x) e^{-0.5t}, \quad (56)$$

$$u_e^*(x, t) = u_l^*(x, t) + 0.5 \partial_t u_l^*(x, t). \quad (57)$$

The source signature is

$$S(x, t) = \frac{\partial_t u_e^* - \kappa_e^* \partial_{xx} u_e^* + G^*(u_e^* - u_l^*)}{I_0^*}. \quad (58)$$

Initial conditions, Dirichlet boundary conditions at $x = 0$ and $x = 1$, and observations are sampled from (u_e^*, u_l^*) .

Appendix B. Training, Validity, and Diagnostic Rules

B.1 Shared Training Protocol

Table S1 summarizes the benchmark protocol. For the main Burgers, coupled reaction–diffusion, and two-temperature chain, the network is a tanh MLP with width 24 and depth 3. Adam uses learning rate 2×10^{-3} for 600 epochs. Training samples $n_{\text{PDE}} = 64$, $n_{\text{data}} = 96$, $n_{\text{IC}} = 48$, and $n_{\text{BC}} = 48$ per epoch. Interior PDE points and data points are sampled uniformly from the corresponding space–time domain; initial-condition points are sampled uniformly in x at $t = 0$; boundary points are sampled uniformly in t at $x = 0$ and $x = 1$. The sampler is seed controlled and resamples the main-chain training points at every epoch. For each checkpoint, the diagnostic sample is drawn once from the same uniform distributions with a fixed seed offset. The source-amplitude positive controls use uniform samples over the same residual-block domains, with a fixed training sample for the Adam/LBFGS run. The block weights used in the training loss and SAEPS weighted residual are $w_{\text{PDE}} = 5$, $w_{\text{data}} = 10$, $w_{\text{IC}} = 10$, and $w_{\text{BC}} = 2$ for the main chain. The main validation uses noise level $\sigma = 0$ and observation fraction 1.0.

Table S1: Benchmark protocol table. Source-code provenance is listed separately in the data and code trace section.

Benchmark	State equation / reference	Target coordinates	coordin-	Training and diagnostic samples	Reference construction
Scalar source amplitude	Closed-form manufactured on $[0, 1]^2$	u^*	$\log \lambda$	Train: 256/512/128/128; diagnostic base: 1/4/2/2 with multipliers 1,4,16,64,256	Closed-form state and forcing
Coupled source amplitude	Closed-form manufactured on $[0, 1]^2$	(u^*, v^*)	$\log \lambda_u, \log \lambda_v$	Train: 256/512/128/128; diagnostic base: 1/4/2/2 with multipliers 1,4,16,64,256	Closed-form two-channel state and forcing
Burgers	Explicit finite-difference reference, 128 $\times$ 2001 grid	periodic	$\log \mu$	Train: 64/96/48/48; diagnostic: 16/24/16/16	Periodic finite-difference reference and interpolation
Coupled reaction-diffusion	Closed-form manufactured on $[0, 1] \times [0, 0.4]$	(u^*, v^*)	$\log a, \log b$	Train: 64/96/48/48; diagnostic: 16/24/16/16	Closed-form state and manufactured sources
Two-temperature	Closed-form manufactured on $[0, 1] \times [0, 0.4]$	(u_e^*, u_l^*)	$\log I_0, \log G$	$\log \kappa_e$, Train: 64/96/48/48; diagnostic: 16/24/16/16	Closed-form state and normalized source signature

B.2 Checkpoint Validity Gate

For the main three-benchmark chain, the evaluation routine applies a checkpoint-quality gate before target-level SAEPS aggregation:

$$\text{PASS} = \mathbf{1}\{E_{\text{PDE}} \leq 5 \times 10^{-2} \wedge E_{\text{data}} \leq 5 \times 10^{-2} \wedge E_{\text{IC}} \leq 5 \times 10^{-2} \wedge E_{\text{BC}} \leq 5 \times 10^{-2} \wedge \text{nRMSE}_{\text{state}} \leq 5 \times 10^{-1}\}. \quad (59)$$

The gate MSEs are computed on unweighted residual blocks. The reported SAEPS quantities are computed on the weighted residual vector. State nRMSE is computed on a 12×8 mesh for the main chain as

$$\text{nRMSE}_{\text{state}} = \frac{\|u_\theta - u^*\|_2}{\|u^*\|_2 + 10^{-12}}, \quad (60)$$

with all state channels concatenated by the tensor norm. Positive-control checkpoints use the stricter state threshold 0.02 and additionally require finite parameter errors. This checkpoint-quality gate is separate from the SAEPS diagnostic solve gate described in Appendix B.3. A target-level row is admissible for aggregation only when the checkpoint gate passes and all conjugate-gradient solves required to apply A_θ for that target-level diagnostic satisfy the relative-residual tolerance.

Table S2 summarizes the validity counts for the main three-benchmark chain. Table S3 reports the corresponding row-level residual and state-error values.

Table S2: Checkpoint validity summary for the main three-benchmark chain.

Benchmark	Target	Rows	PASS	FAIL
Burgers	$\log \mu$	5	4	1
Coupled reaction–diffusion	$\log a$	5	5	0
Coupled reaction–diffusion	$\log b$	5	5	0
Two-temperature	$\log G$	5	5	0
Two-temperature	$\log I_0$	5	5	0
Two-temperature	$\log \kappa_e$	5	5	0

Table S3: Row-level checkpoint validity values for the main three-benchmark chain. Abbreviations: B, Burgers; CRD, coupled reaction–diffusion; 2T, two-temperature.

Benchmark	Seed	Target	Gate	PDE MSE	Data MSE	IC MSE	BC MSE	State nRMSE
B	0	$\log \mu$	PASS	3.06e-02	2.15e-03	1.26e-02	1.91e-02	1.44e-01
B	1	$\log \mu$	PASS	1.81e-02	3.61e-03	1.54e-02	7.47e-03	1.58e-01
B	2	$\log \mu$	PASS	5.43e-03	8.02e-03	1.21e-02	3.48e-02	1.71e-01
B	3	$\log \mu$	PASS	1.24e-02	4.42e-03	7.06e-03	1.29e-02	1.57e-01
B	4	$\log \mu$	FAIL	5.41e-02	4.88e-03	8.81e-03	2.49e-02	1.60e-01
CRD	0	$\log a$	PASS	2.11e-04	2.56e-05	5.48e-05	1.12e-04	7.56e-03
CRD	0	$\log b$	PASS	2.11e-04	2.56e-05	5.48e-05	1.12e-04	7.56e-03
CRD	1	$\log a$	PASS	3.84e-04	5.62e-05	7.07e-05	2.96e-04	1.14e-02
CRD	1	$\log b$	PASS	3.84e-04	5.62e-05	7.07e-05	2.96e-04	1.14e-02
CRD	2	$\log a$	PASS	4.36e-04	9.01e-05	1.75e-04	3.92e-04	1.52e-02
CRD	2	$\log b$	PASS	4.36e-04	9.01e-05	1.75e-04	3.92e-04	1.52e-02
CRD	3	$\log a$	PASS	4.73e-04	7.86e-05	7.50e-05	2.56e-04	1.13e-02
CRD	3	$\log b$	PASS	4.73e-04	7.86e-05	7.50e-05	2.56e-04	1.13e-02
CRD	4	$\log a$	PASS	9.26e-04	6.34e-05	1.31e-04	3.85e-04	1.45e-02
CRD	4	$\log b$	PASS	9.26e-04	6.34e-05	1.31e-04	3.85e-04	1.45e-02
2T	0	$\log G$	PASS	3.58e-05	4.58e-06	3.28e-06	1.24e-05	5.14e-03
2T	0	$\log I_0$	PASS	3.58e-05	4.58e-06	3.28e-06	1.24e-05	5.14e-03
2T	0	$\log \kappa_e$	PASS	3.58e-05	4.58e-06	3.28e-06	1.24e-05	5.14e-03
2T	1	$\log G$	PASS	6.80e-05	5.77e-06	9.51e-06	2.04e-05	5.90e-03
2T	1	$\log I_0$	PASS	6.80e-05	5.77e-06	9.51e-06	2.04e-05	5.90e-03
2T	1	$\log \kappa_e$	PASS	6.80e-05	5.77e-06	9.51e-06	2.04e-05	5.90e-03
2T	2	$\log G$	PASS	8.25e-05	3.57e-05	5.26e-05	1.06e-04	1.43e-02

Benchmark	Seed	Target	Gate	PDE MSE	Data MSE	IC MSE	BC MSE	State nRMSE
2T	2	$\log I_0$	PASS	8.25e-05	3.57e-05	5.26e-05	1.06e-04	1.43e-02
2T	2	$\log \kappa_e$	PASS	8.25e-05	3.57e-05	5.26e-05	1.06e-04	1.43e-02
2T	3	$\log G$	PASS	9.79e-05	5.67e-06	1.14e-05	2.08e-05	6.22e-03
2T	3	$\log I_0$	PASS	9.79e-05	5.67e-06	1.14e-05	2.08e-05	6.22e-03
2T	3	$\log \kappa_e$	PASS	9.79e-05	5.67e-06	1.14e-05	2.08e-05	6.22e-03
2T	4	$\log G$	PASS	1.40e-04	5.81e-06	1.18e-05	2.11e-05	6.59e-03
2T	4	$\log I_0$	PASS	1.40e-04	5.81e-06	1.18e-05	2.11e-05	6.59e-03
2T	4	$\log \kappa_e$	PASS	1.40e-04	5.81e-06	1.18e-05	2.11e-05	6.59e-03

B.3 SAEPS Linear Solve and Label Rules

The main chain computes

$$\gamma = \gamma_\alpha \lambda_{\max}(J_\theta^\top J_\theta), \quad \gamma_\alpha = 10^{-7} \quad (61)$$

for core target-level diagnostics. The damping-consistency audit uses the extended grid $\gamma_\alpha \in \{10^{-12}, 10^{-10}, 10^{-8}, 10^{-6}, 10^{-4}, 10^{-2}, 1, 10^2\}$. The state-elimination operator is applied as

$$A_\theta y = y - J_\theta z, \quad (J_\theta^\top J_\theta + \gamma I)z = J_\theta^\top y. \quad (62)$$

The conjugate-gradient solve is initialized at zero and is marked converged when the relative residual recorded by the implementation satisfies

$$\frac{\|(J_\theta^\top J_\theta + \gamma I)z - J_\theta^\top y\|_2}{\|J_\theta^\top y\|_2 + 10^{-30}} \leq \tau_{\text{CG}}. \quad (63)$$

The tolerance and maximum iteration count are $\tau_{\text{CG}} = 10^{-8}$ and 300 for the main chain, and $\tau_{\text{CG}} = 10^{-8}$ and 500 for the source-amplitude positive controls. The solve is run for each column of J_λ and once for the weighted residual vector; target-level aggregation requires all corresponding solve-convergence checks to pass.

The score-alignment quantity is

$$\text{align} = \frac{\langle \Delta \lambda^{\text{se}}, \lambda_{\text{ref}} - \lambda_{\text{ckpt}} \rangle}{\|\Delta \lambda^{\text{se}}\|_2 \|\lambda_{\text{ref}} - \lambda_{\text{ckpt}}\|_2 + 10^{-30}}, \quad (64)$$

where $\Delta \lambda^{\text{se}}$ is the SAEPS diagnostic step, λ_{ref} is the manufactured or synthetic reference parameter, and λ_{ckpt} is the checkpoint parameter. This quantity is a manufactured/synthetic validation diagnostic only, because it requires the reference parameter and is therefore not deployable in real inverse problems without known truth.

The weak-eigendirection value is the smallest eigenvalue of the symmetrized state-eliminated curvature matrix,

$$\lambda_{\min}^{\text{weak}} = \lambda_{\min} \left(\frac{F_\lambda^{\text{se}} + (F_\lambda^{\text{se}})^\top}{2} \right). \quad (65)$$

The same function also returns the projection of the parameter-error direction onto the corresponding weakest eigenvector, but the current main-chain label rule uses the eigenvalue threshold rather than that projection ratio.

The main-chain internal decision rule applies the following priority order:

$$\begin{cases} \textit{score-misaligned}, & \text{align} < 0, \\ \textit{state-absorbed}, & \text{align} \geq 0 \text{ and } \eta^{\text{se}} < 0.15, \\ \textit{weak-eigendirection}, & \text{align} \geq 0, \eta^{\text{se}} \geq 0.15, \lambda_{\min}^{\text{weak}} < 10^{-4}, \\ \textit{locally reliable}, & \text{otherwise.} \end{cases} \quad (66)$$

The source-amplitude positive-control decision rule applies:

$$\begin{cases} \textit{unstable-denominator}, & \text{denominator unstable}, \\ \textit{low-retained-sensitivity}, & \eta^{\text{se}} \leq 0.2, \\ \textit{score-misaligned}, & \eta^{\text{se}} > 0.2 \text{ and } \text{score} \leq 0.5, \\ \textit{reliable-local-signal}, & \text{otherwise.} \end{cases} \quad (67)$$

The paper-facing summary reports *reliable-local-signal*, *state-absorbed*, and *score-misaligned*, with grid, damping, and profile status reported separately.

Appendix C. Supplementary Numerical Tables and Figures

Table S4: Target-level SAEPS numerical summary. Values are medians over validity-passing checkpoints.

Target	Label	n	State nRMSE	Log error	F_{λ}^{raw}	F_{λ}^{se}	η^{se}	Score
Scalar source / $\log \lambda$	reliable	5	2.537e-05	6.758e-06	3.013e+02	2.999e+02	9.952e-01	1.0000
Coupled source / $\log \lambda_u$	reliable	5	2.721e-05	5.950e-06	3.013e+02	2.999e+02	9.958e-01	1.0000
Coupled source / $\log \lambda_v$	reliable	5	2.721e-05	4.213e-06	1.298e+02	1.297e+02	9.997e-01	1.0000
Burgers / $\log \mu$	state-absorbed	4	1.572e-01	2.208e-01	4.277e+04	1.374e+03	3.204e-02	1.0000
CRD / $\log a$	state-absorbed	5	1.139e-02	4.465e-02	1.977e+04	1.353e+03	6.848e-02	0.9994
CRD / $\log b$	state-absorbed	5	1.139e-02	4.443e-02	7.741e+03	5.278e+02	6.812e-02	0.9994
2T / $\log G$	score-misaligned	5	6.216e-03	2.587e-01	6.384e+01	3.028e+00	4.676e-02	-0.1677
2T / $\log I_0$	score-misaligned	5	6.216e-03	3.661e-01	3.999e+02	1.781e+01	4.464e-02	-0.1677
2T / $\log \kappa_e$	score-misaligned	5	6.216e-03	2.616e-01	1.070e+03	4.385e+01	4.169e-02	-0.1677

Table S5: Target-level consistency checks corresponding to Table S4.

Target	Regime	Grid status	Damping/profile status	
Scalar source / $\log \lambda$	Retained local signal	grid-stable	plateau-stable; checkpoint-centered	
Coupled source / $\log \lambda_u$	Retained local signal	grid-stable	plateau-stable; checkpoint-centered	
Coupled source / $\log \lambda_v$	Retained local signal	grid-stable	plateau-stable; checkpoint-centered	
Burgers / $\log \mu$	State absorption	grid-stable	damping-sensitive; checkpoint-centered	
Ξ CRD / $\log a$	State absorption	grid-stable	damping-sensitive; checkpoint-centered	
	CRD / $\log b$	grid-stable	damping-sensitive; not recorded	
	2T / $\log G$	grid-sensitive	damping-sensitive; not recorded	
2T / $\log I_0$	Score misalignment	grid-sensitive	damping-sensitive; checkpoint-centered	
2T / $\log \kappa_e$	Score misalignment	grid-sensitive	damping-sensitive; checkpoint-centered	

Table S6: Seed-level retained-sensitivity values used by Supplementary Fig. S1. Abbreviations: S, scalar source amplitude; CS, coupled source amplitude; B, Burgers; CRD, coupled reaction–diffusion; 2T, two-temperature.

Benchmark	Seed	Target	η^{se}
S	0	$\log \lambda$	9.952e-01
S	1	$\log \lambda$	9.952e-01
S	2	$\log \lambda$	9.952e-01
S	3	$\log \lambda$	9.952e-01
S	4	$\log \lambda$	9.952e-01
CS	0	$\log \lambda_u$	9.958e-01
CS	1	$\log \lambda_u$	9.958e-01
CS	2	$\log \lambda_u$	9.958e-01
CS	3	$\log \lambda_u$	9.958e-01
CS	4	$\log \lambda_u$	9.958e-01
CS	0	$\log \lambda_v$	9.997e-01
CS	1	$\log \lambda_v$	9.997e-01
CS	2	$\log \lambda_v$	9.997e-01
CS	3	$\log \lambda_v$	9.997e-01
CS	4	$\log \lambda_v$	9.997e-01
B	0	$\log \mu$	5.590e-03
B	1	$\log \mu$	5.086e-03
B	2	$\log \mu$	6.960e-03
B	3	$\log \mu$	2.320e-03
CRD	0	$\log a$	7.539e-02
CRD	1	$\log a$	5.403e-02
CRD	2	$\log a$	5.923e-02
CRD	3	$\log a$	4.654e-02

Benchmark	Seed	Target	η^{se}
CRD	4	$\log a$	7.640e-02
CRD	0	$\log b$	7.310e-02
CRD	1	$\log b$	4.715e-02
CRD	2	$\log b$	4.990e-02
CRD	3	$\log b$	5.457e-02
CRD	4	$\log b$	7.513e-02
2T	0	$\log G$	5.332e-02
2T	1	$\log G$	4.828e-02
2T	2	$\log G$	4.109e-02
2T	3	$\log G$	2.695e-02
2T	4	$\log G$	6.149e-02
2T	0	$\log I_0$	4.814e-02
2T	1	$\log I_0$	4.587e-02
2T	2	$\log I_0$	4.897e-02
2T	3	$\log I_0$	2.574e-02
2T	4	$\log I_0$	5.151e-02
2T	0	$\log \kappa_e$	4.673e-02
2T	1	$\log \kappa_e$	4.312e-02
2T	2	$\log \kappa_e$	4.162e-02
2T	3	$\log \kappa_e$	2.498e-02
2T	4	$\log \kappa_e$	4.890e-02

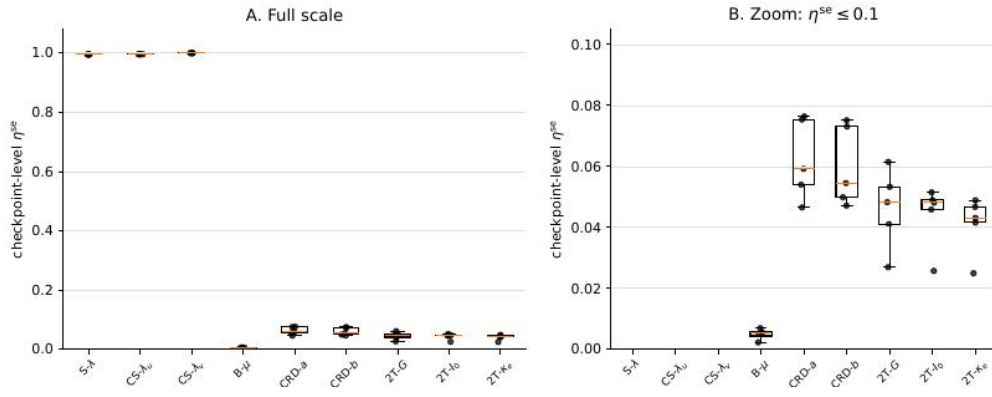


Figure S1: Seed-level retained sensitivity distributions for the target rows in Table S6.

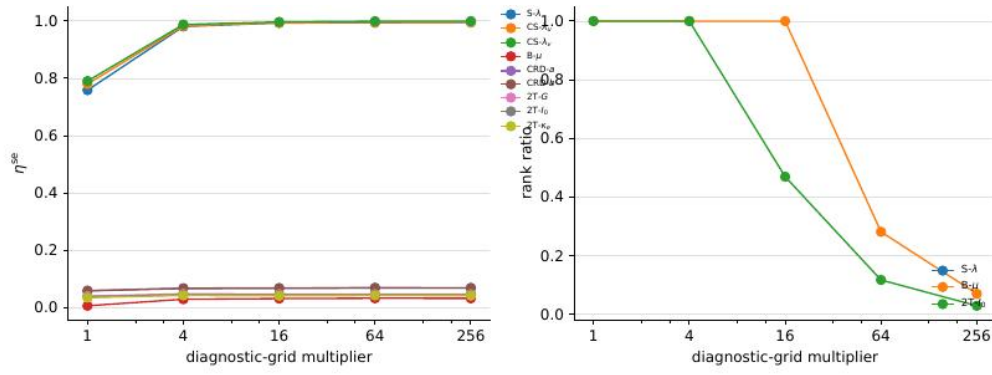


Figure S2: Full diagnostic-grid and rank audit exported for the current paper package.

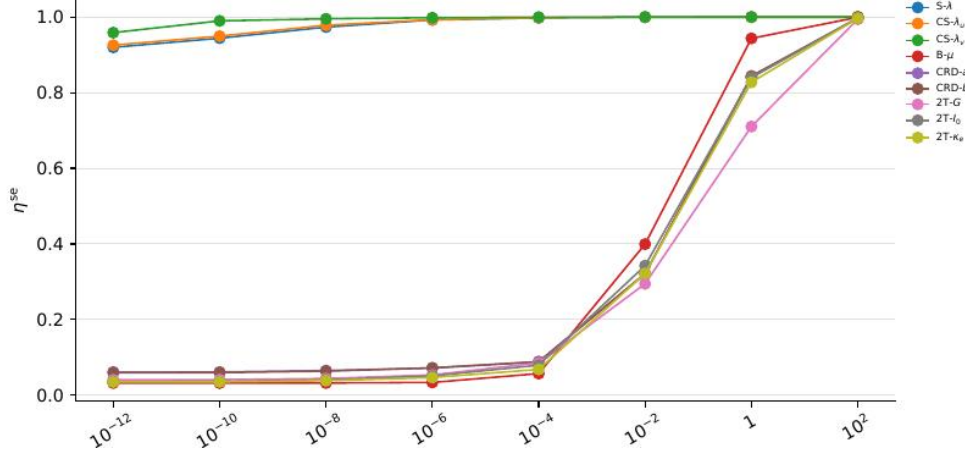


Figure S3: Full damping sweep used in the current figure package. The target-level medians are evaluated at the core scale $\gamma_\alpha = 10^{-7}$; the sweep shown here is the extended consistency audit over $\gamma_\alpha \in \{10^{-12}, 10^{-10}, 10^{-8}, 10^{-6}, 10^{-4}, 10^{-2}, 1, 10^2\}$.

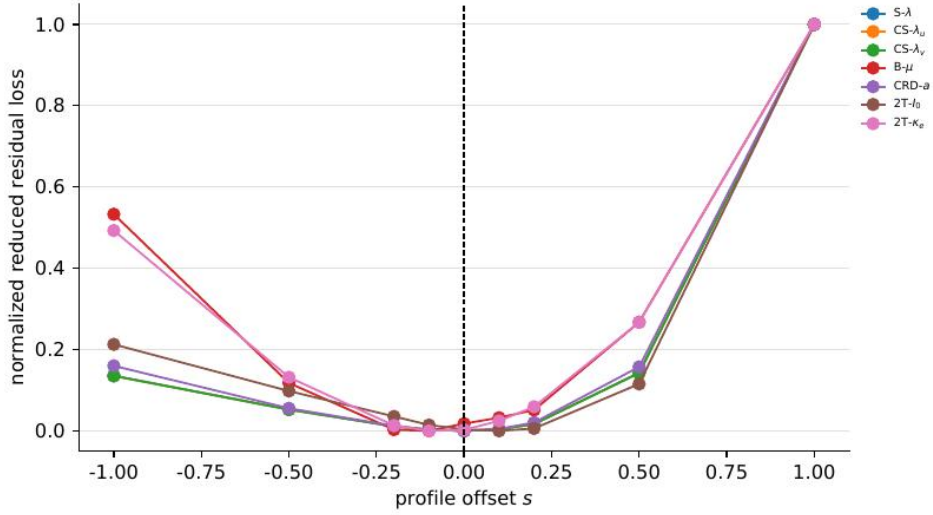


Figure S4: Full reduced-profile audit traces exported for the current paper package.

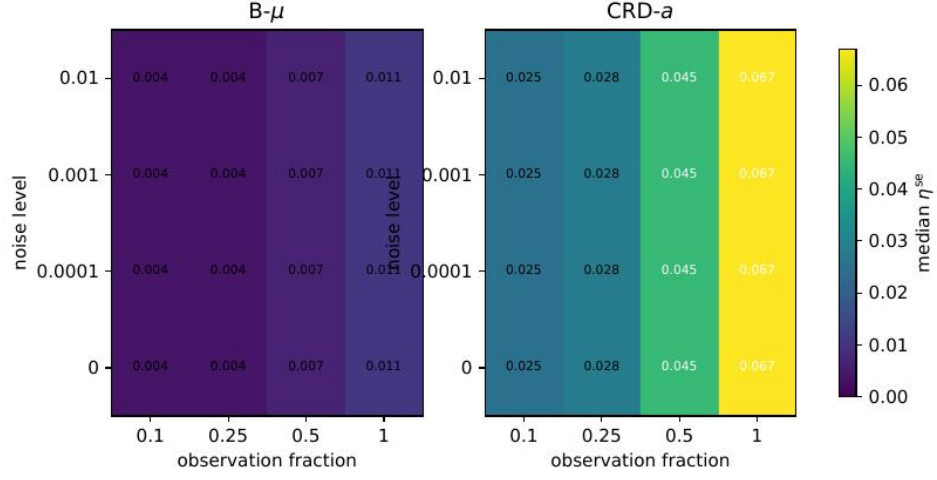


Figure S5: Selected noise and sparsity stress tests. These rows are scope-limited stress tests and do not constitute a full Cartesian robustness claim.

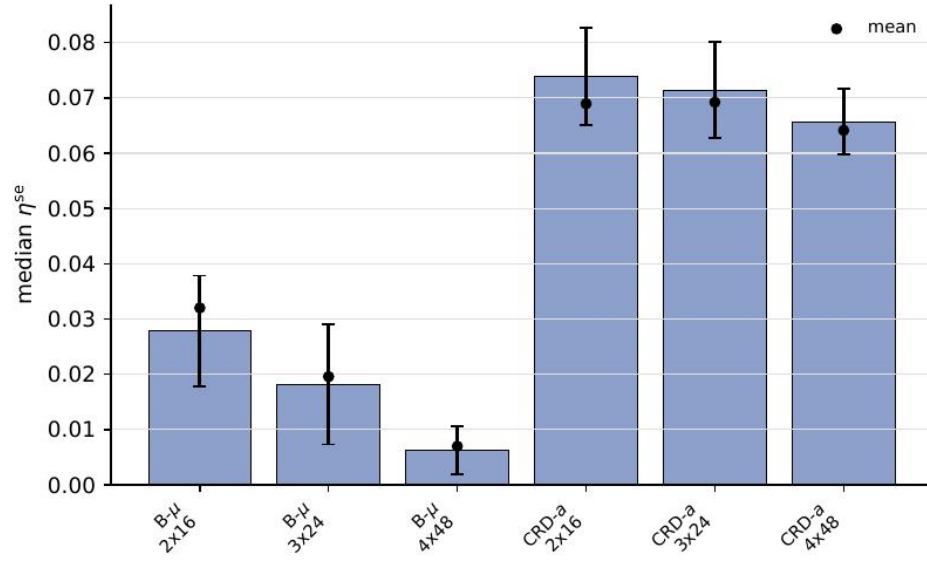


Figure S6: Selected architecture-dependence audit exported for the current paper package.

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